

|          |                                                                      |           |
|----------|----------------------------------------------------------------------|-----------|
|          |                                                                      |           |
| 8(a)(i)  | <i>Either</i> Newton's third law <i>or</i> equal and opposite forces | <b>B1</b> |
|          | force on magnet is upwards                                           | <b>B1</b> |
|          | so force on wire downwards                                           | <b>B1</b> |
| 8(a)(ii) | using (Fleming's) left-hand rule                                     | <b>M1</b> |
|          | current from B to A                                                  | <b>A1</b> |
| 8(b)     | sinusoidal wave with at least 1 cycle                                | <b>B1</b> |
|          | peaks at +6.4 mN and –6.4 mN                                         | <b>B1</b> |
|          | time period 25 ms                                                    | <b>B1</b> |

| Question | Answer                                                                                                               | Marks     |
|----------|----------------------------------------------------------------------------------------------------------------------|-----------|
| 9        | X-rays (are used)                                                                                                    | <b>B1</b> |
|          | (object is) scanned in sections / slices                                                                             | <b>B1</b> |
|          | <i>either:</i> scans taken at many angles / directions<br><i>or</i> images of each section / slice are 2-dimensional | <b>B1</b> |
|          | scans of many sections / slices are combined                                                                         | <b>B1</b> |
|          | (to give) 3-dimensional image (of whole structure)                                                                   | <b>B1</b> |

| Question | Answer | Marks |
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